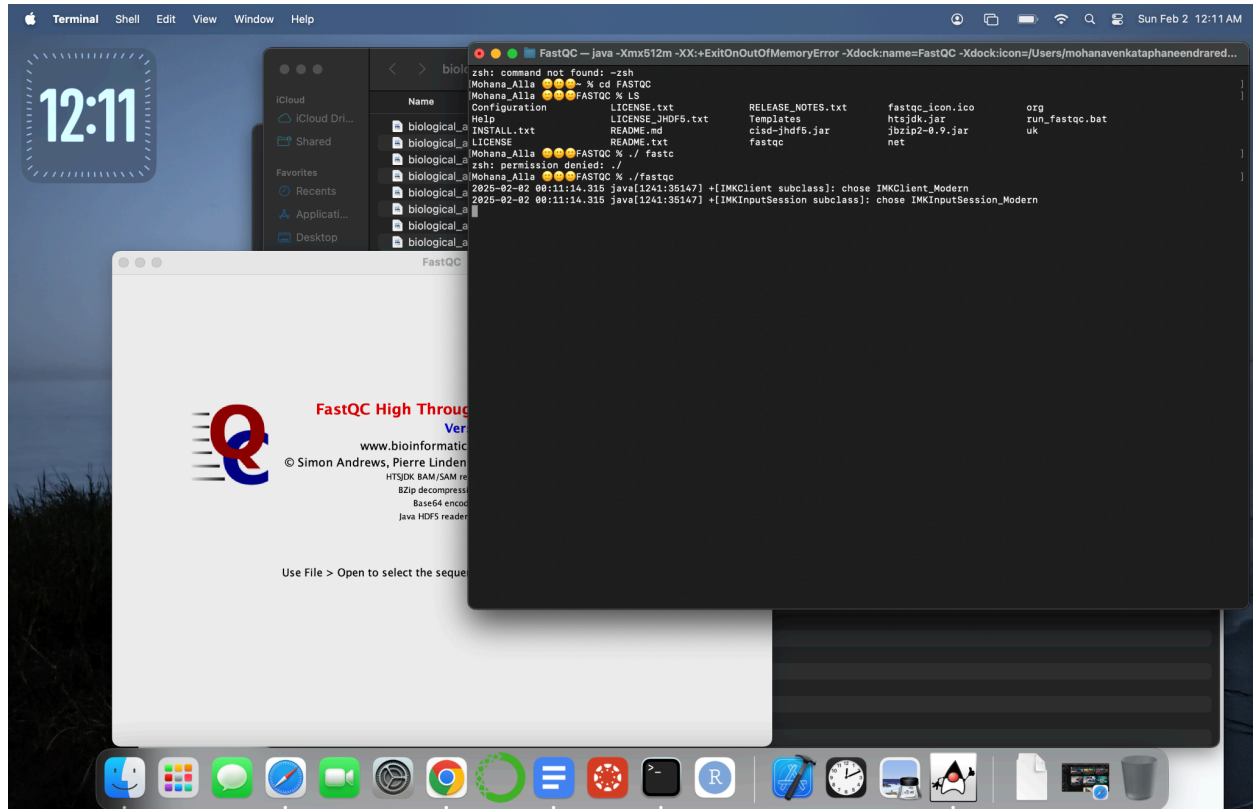


Homework_1

FastQC Report Summary for ENCFF002DKA.fastq



1. Basic Statistics

Thu 30 Jan 2025
ENCFF002DKA.fastq

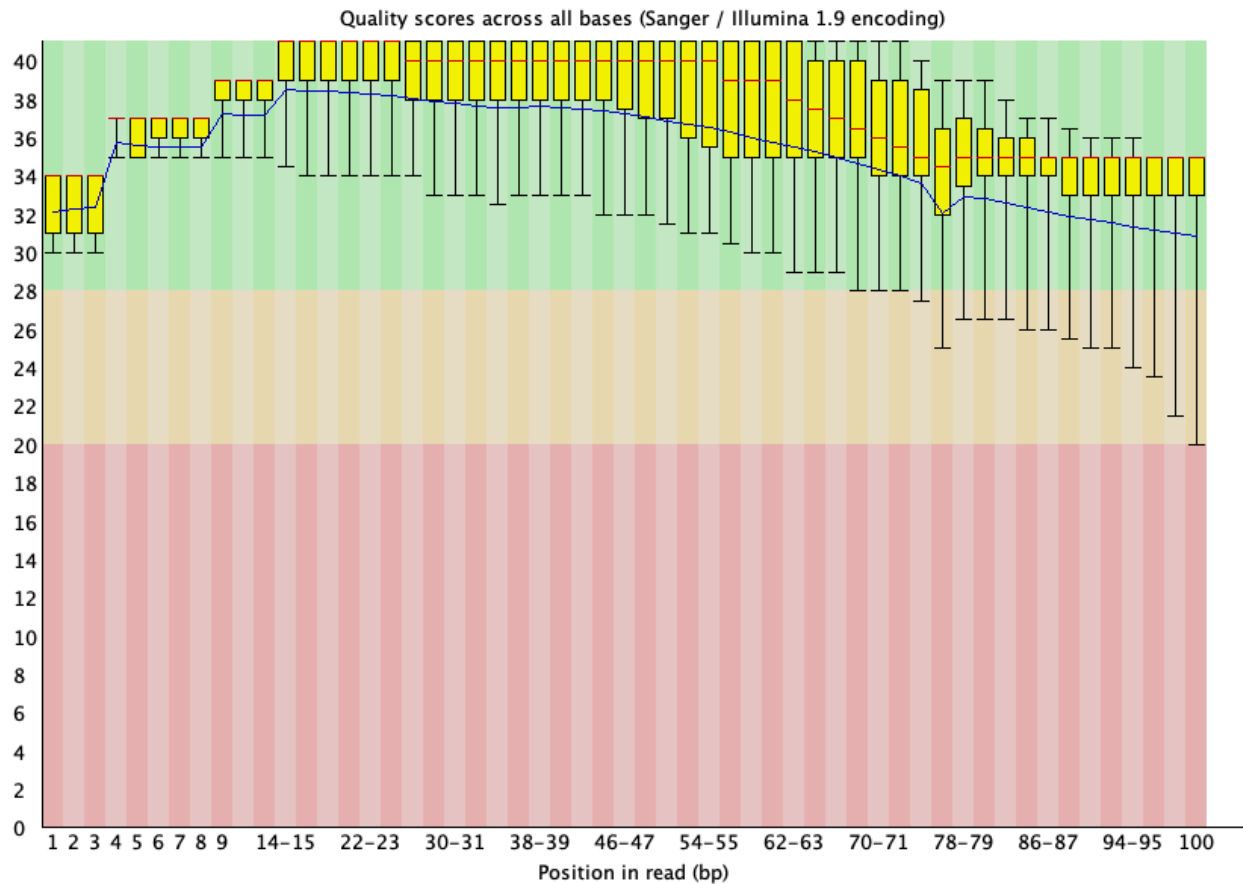
✓ Basic Statistics

Measure	Value
Filename	ENCFF002DKA.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	74096905
Total Bases	7.4 Gbp
Sequences flagged as poor quality	0
Sequence length	100
%GC	47

- Provides general information about the sequence file, including:
 - File name and type

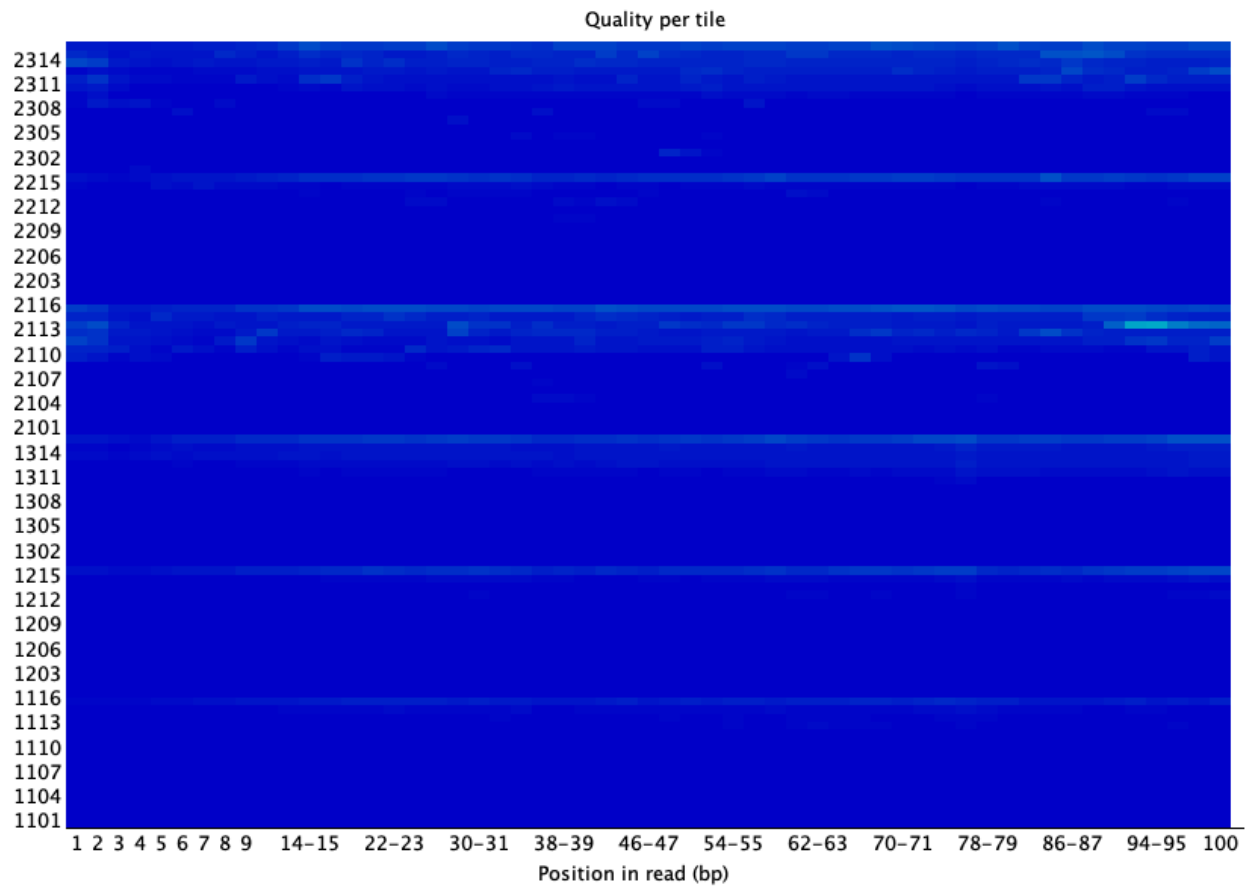
- Encoding format of quality scores
- Total number of sequences
- Filtered sequences
- Sequence length range
- GC content percentage
- **Status:** PASS (No issues detected)

2. Per Base Sequence Quality



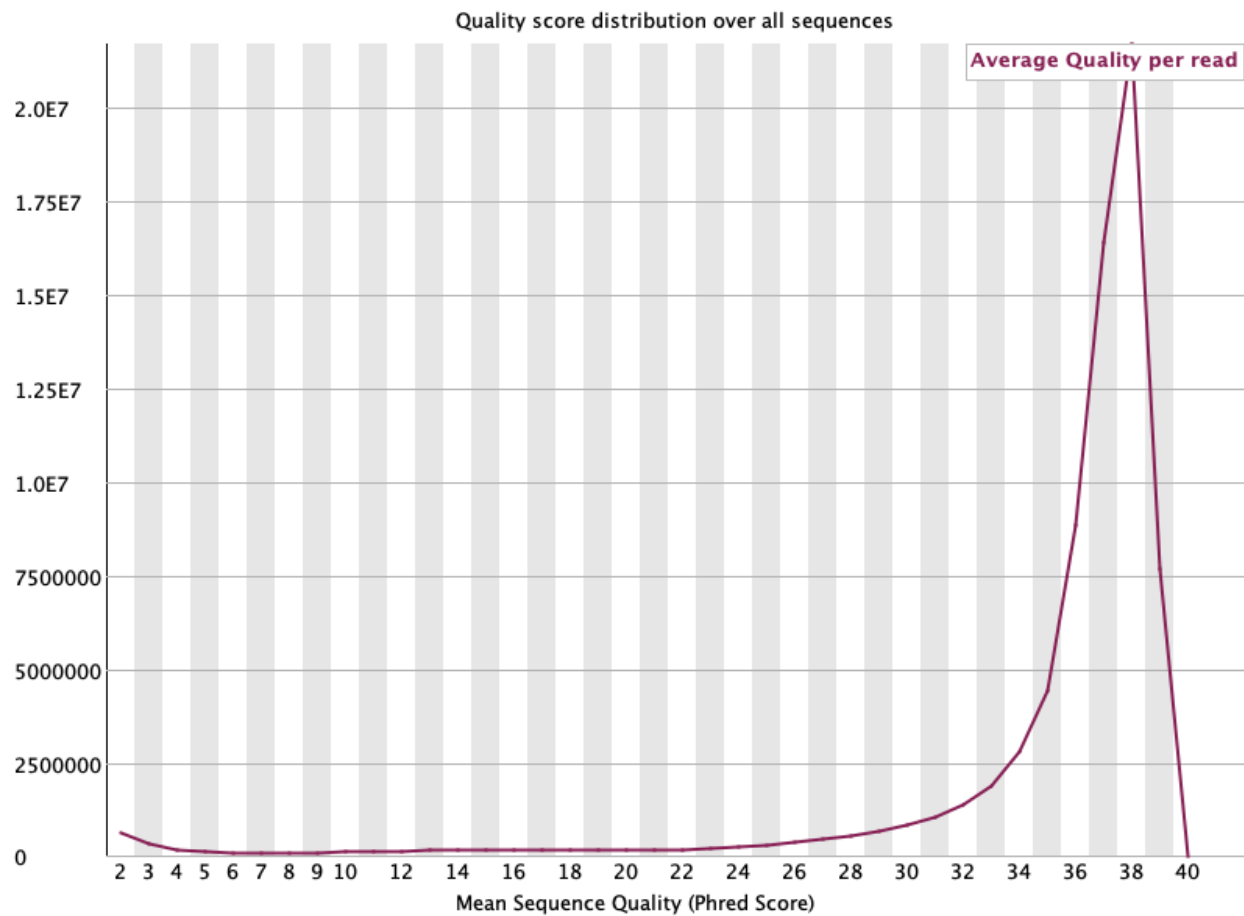
- Box plot showing the quality distribution at each base position.
- Median (red line), interquartile range (yellow box), and mean (blue line) are displayed.
- Background color-coded for quality:
 - Green: High quality
 - Orange: Moderate quality
 - Red: Poor quality
- **Status:** PASS (Good quality across bases)

3. Per Tile Sequence Quality



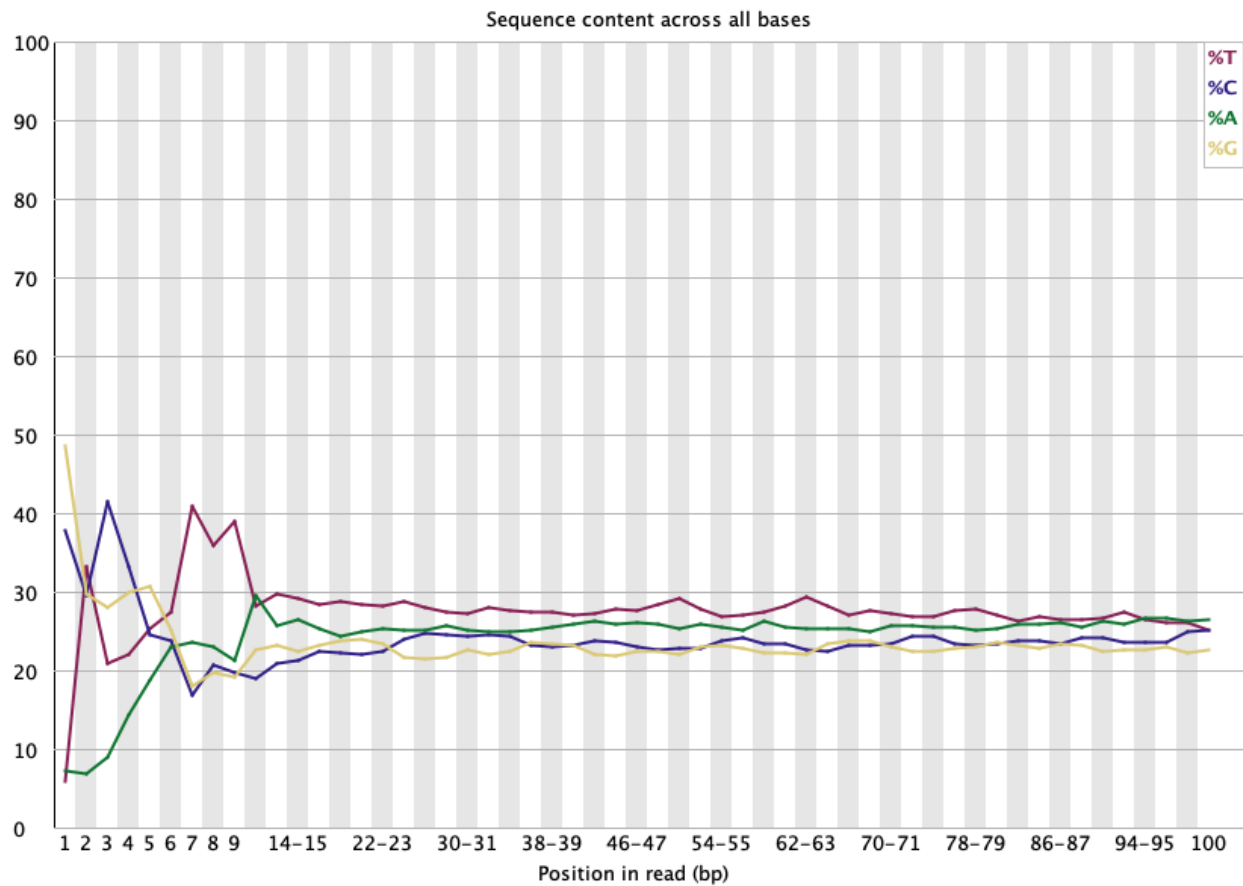
- Identifies potential systematic errors in sequencing tiles.
- Cold colors (blue) indicate average or high quality, while hot colors (red/yellow) indicate poor quality.
- Common issues include bubbles, smudges, or debris in the flowcell.
- **Status:** PASS (No major tile-specific quality drops)

4. Per Sequence Quality Scores



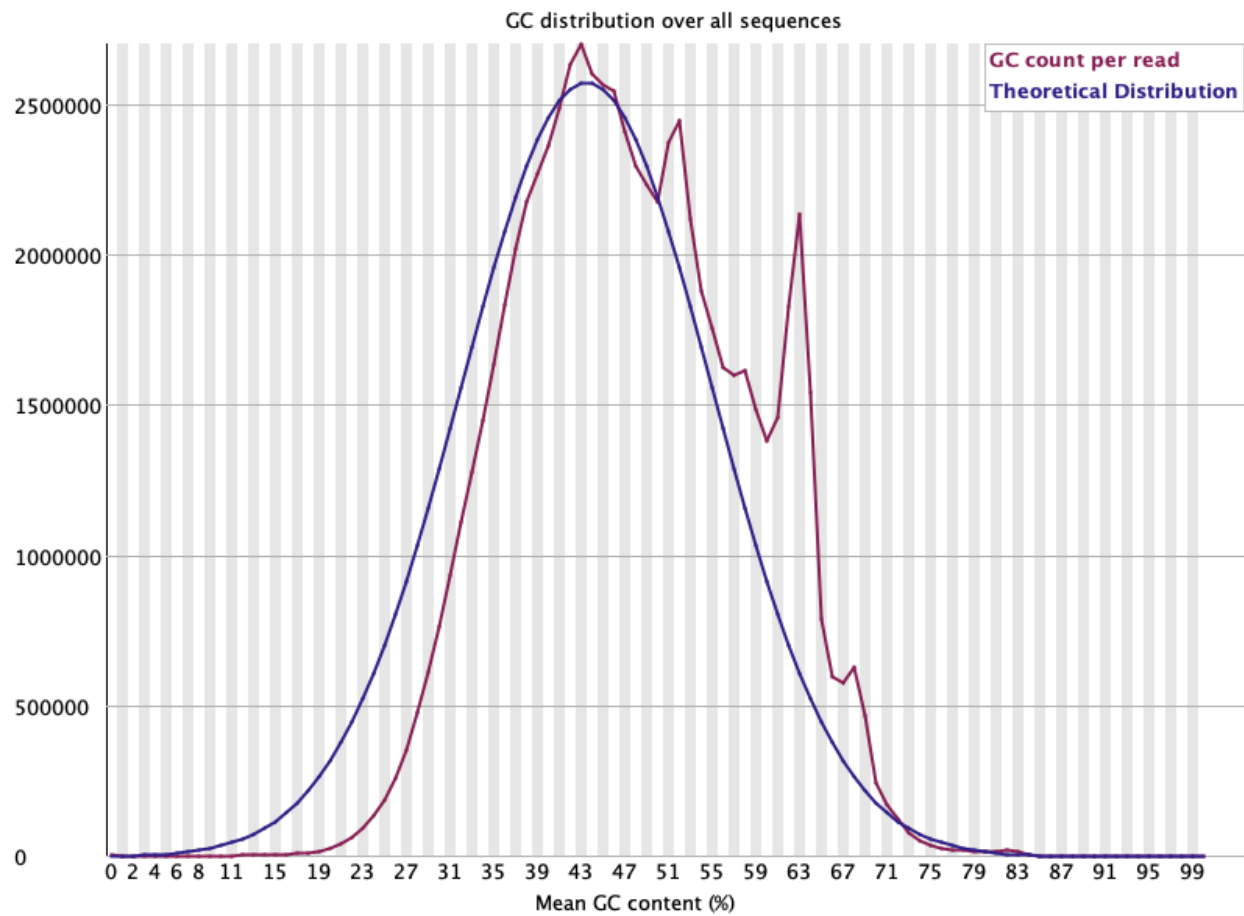
- Histogram of average quality scores per sequence.
- Detects subsets of sequences with consistently low quality.
- **Status:** PASS (No significant proportion of low-quality reads)

5. Per Base Sequence Content



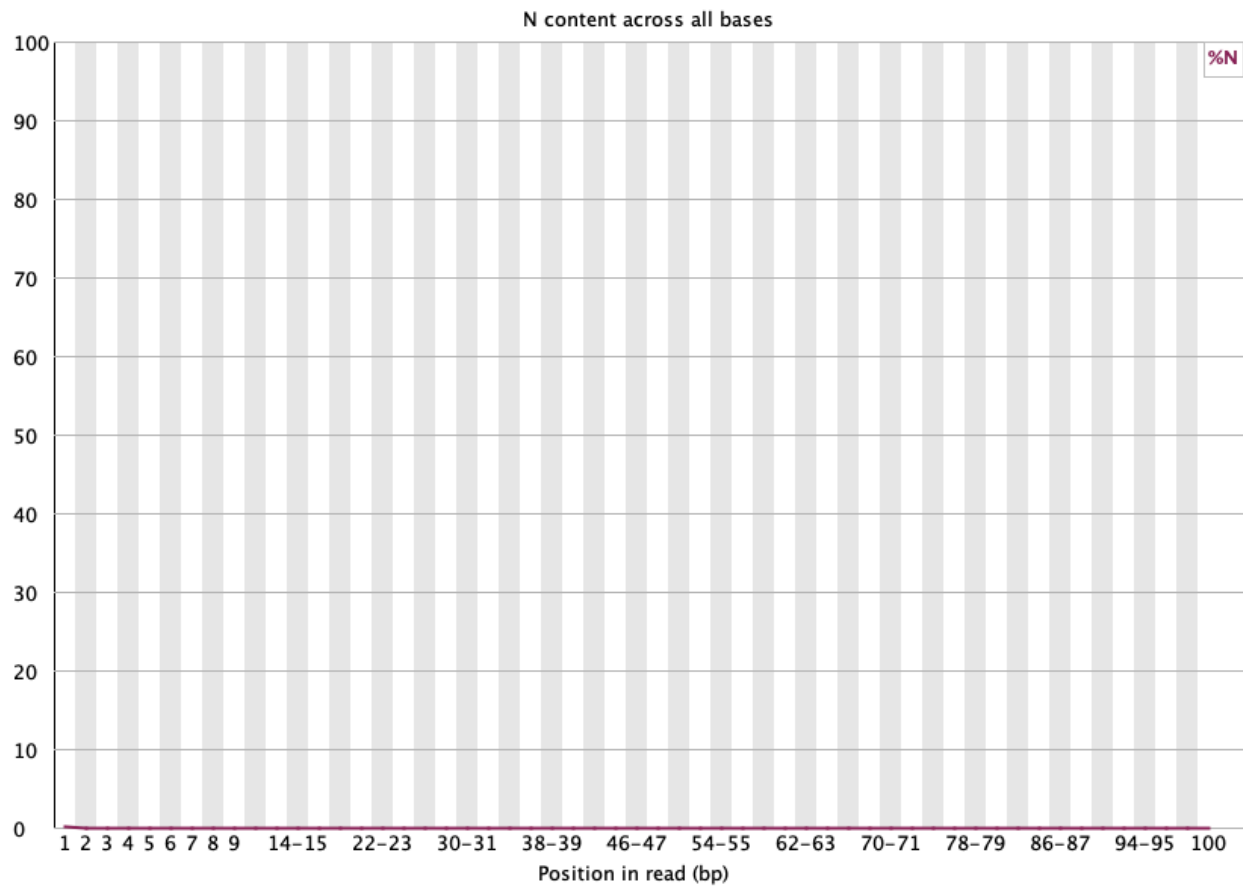
- Tracks nucleotide proportions (A, T, G, C) at each position.
- Ideally, proportions should be consistent, except for known biases (e.g., RNA-Seq).
- **Status:** FAIL (Significant bias detected, indicating possible adapter contamination or sequence composition issues)

6. Per Sequence GC Content



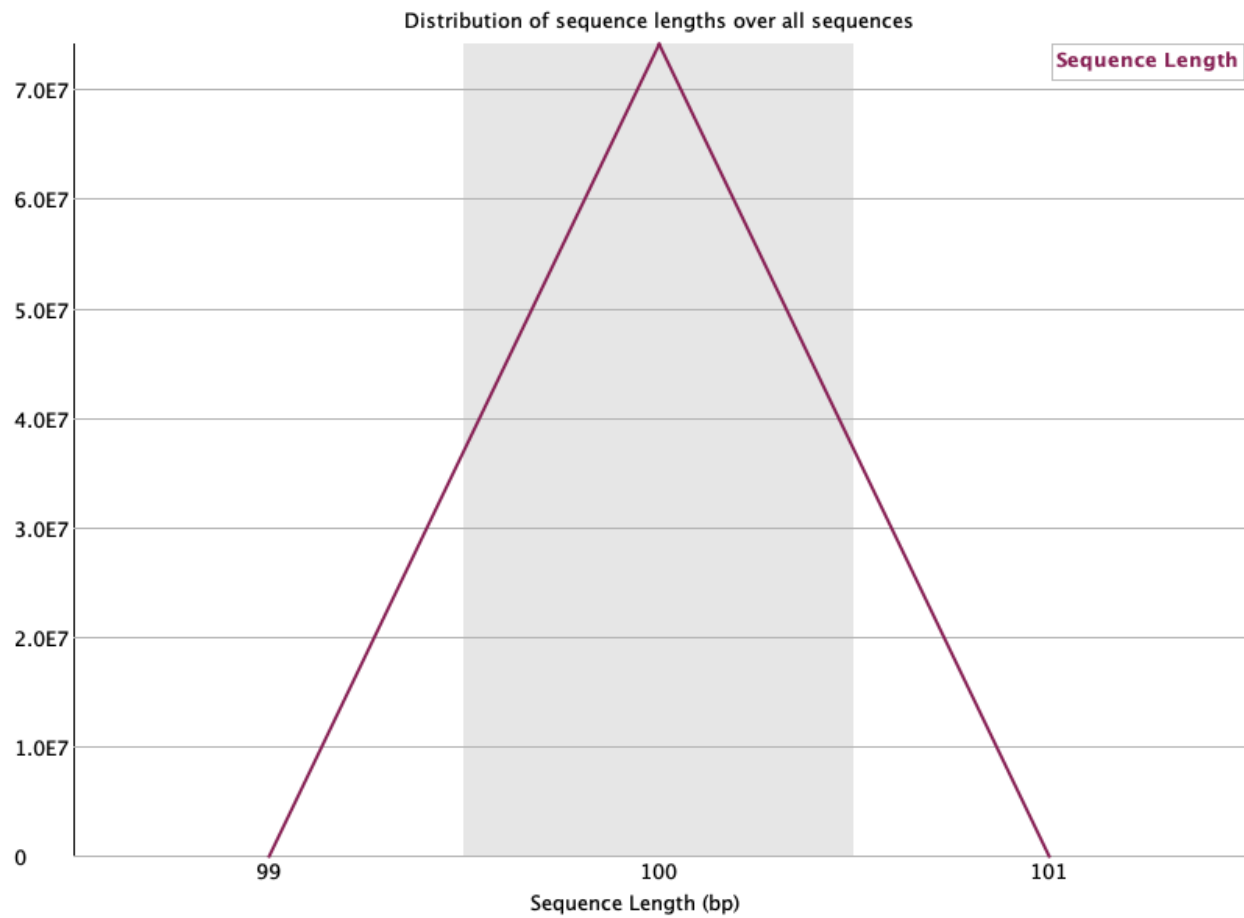
- Compares GC content distribution to an expected normal distribution.
- Large deviations may indicate contamination or unexpected sequencing bias.
- **Status:** WARNING (Deviations observed, suggesting possible contamination)

7. Per Base N Content



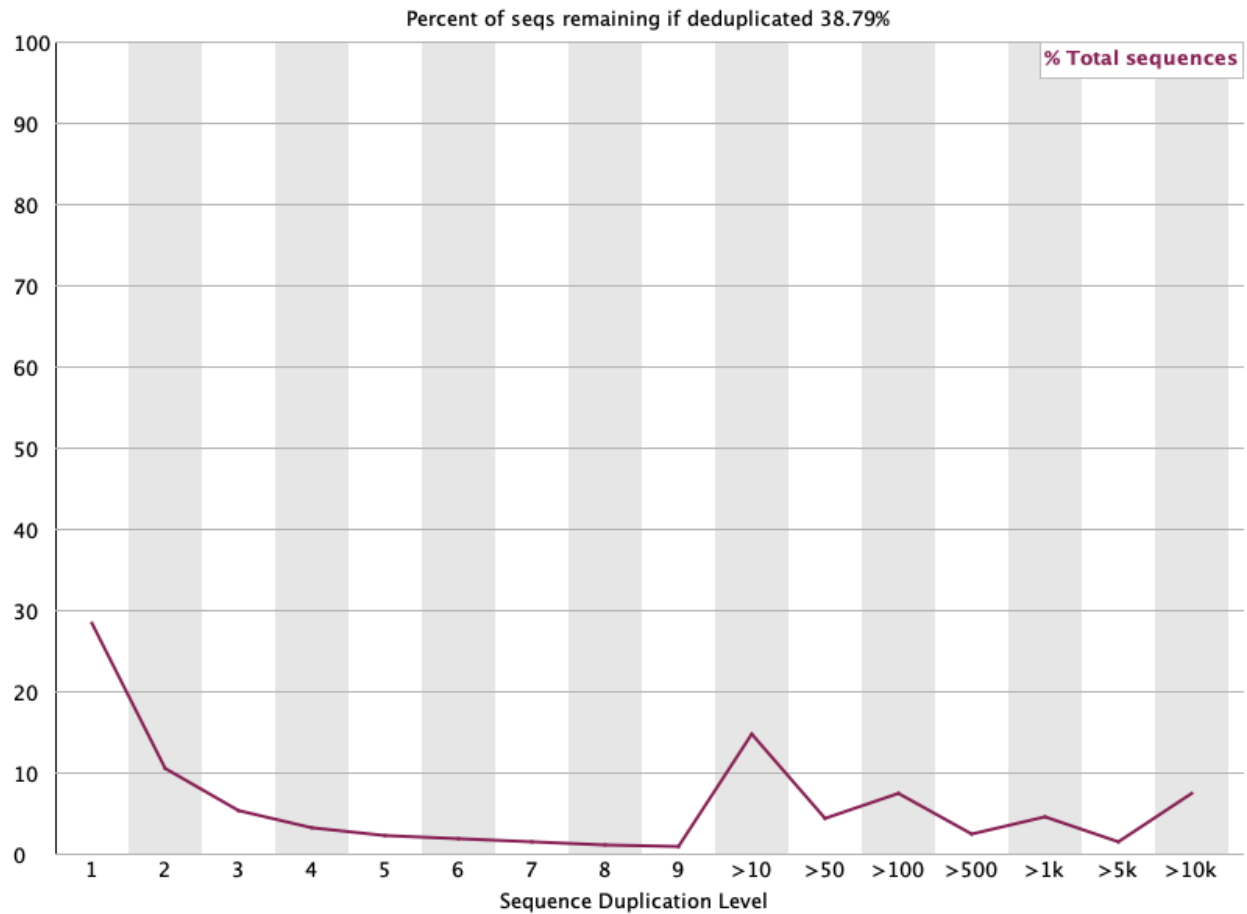
- Detects ambiguous bases (N) across positions.
- High N content suggests poor sequencing quality or failed base calling.
- **Status:** PASS (Minimal N content detected)

8. Sequence Length Distribution



- Shows the distribution of read lengths.
- Uniform read lengths are expected unless trimming or variable-length sequencing is used.
- **Status:** PASS (Expected length distribution)

9. Sequence Duplication Levels



- Measures duplicate sequences within the dataset.
- High duplication may indicate PCR over-amplification or a biased library.
- **Status:** FAIL (High duplication levels detected, requiring further analysis)

10. Overrepresented Sequences

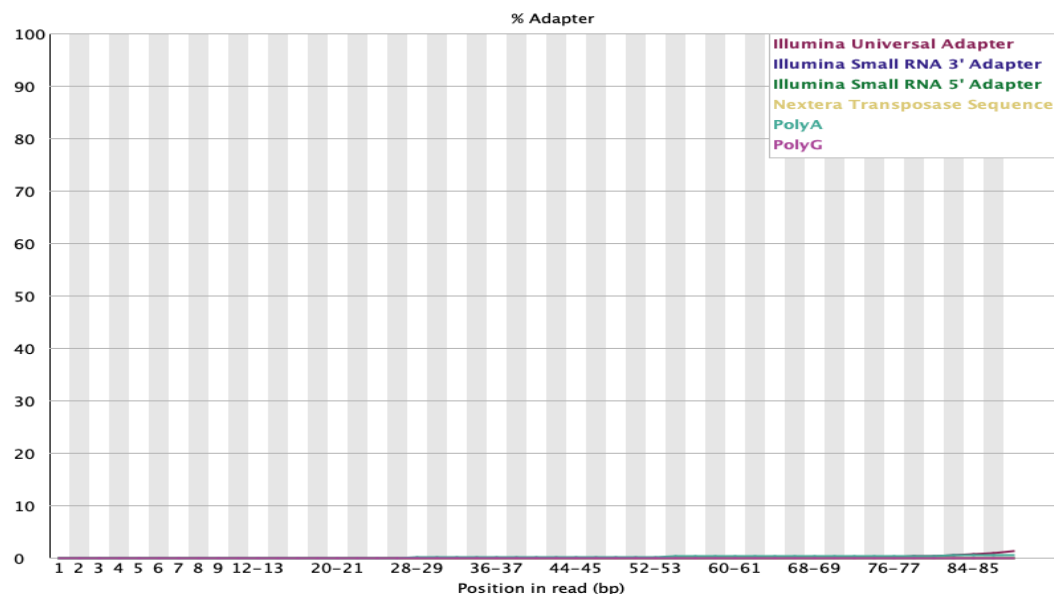
Thu 30 Jan 2025
ENCFF002DKA.fastq

! Overrepresented sequences

Sequence	Count	Percentage	Possible Source
CCCCTCCTTAGGCAACCTGGTGGTCCCCCGCTCCCGGGAGGTCACCATAT	536974	0.7246915373860757	No Hit
CCCTCCTTAGGCAACCTGGTGGTCCCCCGCTCCCGGGAGGTCACCATATT	385177	0.5198287296885072	No Hit
GTCTGGAGTCTTGAAGCTTGACTACCTACGTTCTCTACAAATGGACC	373105	0.5035365512230234	No Hit
CCTCCTTAGGCAACCTGGTGGTCCCCCGCTCCCGGGAGGTCACCATATTG	212328	0.2865544788949012	No Hit
GCTCCGTTCCGACCTGGGCCGGTTCACCCCTCCTTAGGCAACCTGGTGG	183447	0.24757714239211476	No Hit
CTGGAGTCTTGAAGCTTGACTACCTACGTTCTCTACAAATGGACCTT	173670	0.2343822592859985	No Hit
CTCCGTTCCGACCTGGGCCGGTTCACCCCTCCTTAGGCAACCTGGTGGT	141489	0.19095129546908873	No Hit
CACCCCTCCTTAGGCAACCTGGTGGTCCCCCGCTCCCGGGAGGTCACCAT	140437	0.18953153306470763	No Hit
CTCCTTAGGCAACCTGGTGGTCCCCCGCTCCCGGGAGGTCACCATATTGA	128392	0.17327579336815754	No Hit
GTGGCTATTACAGGCGCGATCCCACTACTGATCAGCACGGGAGTTTGA	104768	0.141393220135173	No Hit
GGCAACCTGGTGGTCCCCCGCTCCCGGGAGGTCACCATATTGATGCCGAA	92465	0.12478928775770054	No Hit
GCCAGGCTGGAGTGCAGTGGCTATTACAGGCGCGATCCCACTACTGAT	87748	0.11842329986657338	No Hit
GGATGTGTCTGGAGTCTTGAAGCTTGACTACCTACGTTCTCTACAAA	81353	0.10979270996541624	No Hit
CTCCTCTATCGGGGATGGTCGTCCTCTTCGACCGAGCGCGAGCTTCGGG	79745	0.10762257883780706	No Hit
GTGGGTATAATACTAAGTTGAGATGATATCATTTACGGGGAAGGCGCTT	78050	0.10533503389918919	No Hit
GCTCAGGCTGGAGTGCAGTGGCTATTACAGGCGCGATCCCACTACTGAT	77431	0.1044996413817824	No Hit
CGGGGATGTCGTCCTCTTCGACCGAGCGCGAGCTTCGGGAGGGACGCA	77131	0.10409476617140756	No Hit

- Identifies sequences occurring at unusually high frequencies.
- May indicate adapter contamination or biological significance.
- **Status:** WARNING (Presence of overrepresented sequences, further investigation needed)

11. Adapter Content



- Checks for adapter contamination.
- Presence of adapter sequences suggests the need for trimming before analysis.
- **Status:** PASS (No significant adapter contamination detected)

Overall Report Summary:

- **PASS:** Basic statistics, per base sequence quality, per tile sequence quality, per sequence quality scores, per base N content, sequence length distribution, and adapter content.
- **WARNING:** Per sequence GC content, overrepresented sequences.
- **FAIL:** Per base sequence content, sequence duplication levels.

Recommendations:

1. Investigate sequence content bias (adapter trimming or sequence-specific anomalies).
2. Examine GC content variations for potential contamination.
3. Assess high sequence duplication levels and their impact on downstream analysis.
4. Consider filtering overrepresented sequences to improve data quality.

This report provides an overview of the quality metrics and highlights potential concerns that may impact downstream analysis.

FastQC Report Summary for ENCFF002DKE.FASTQC

1. Basic Statistics

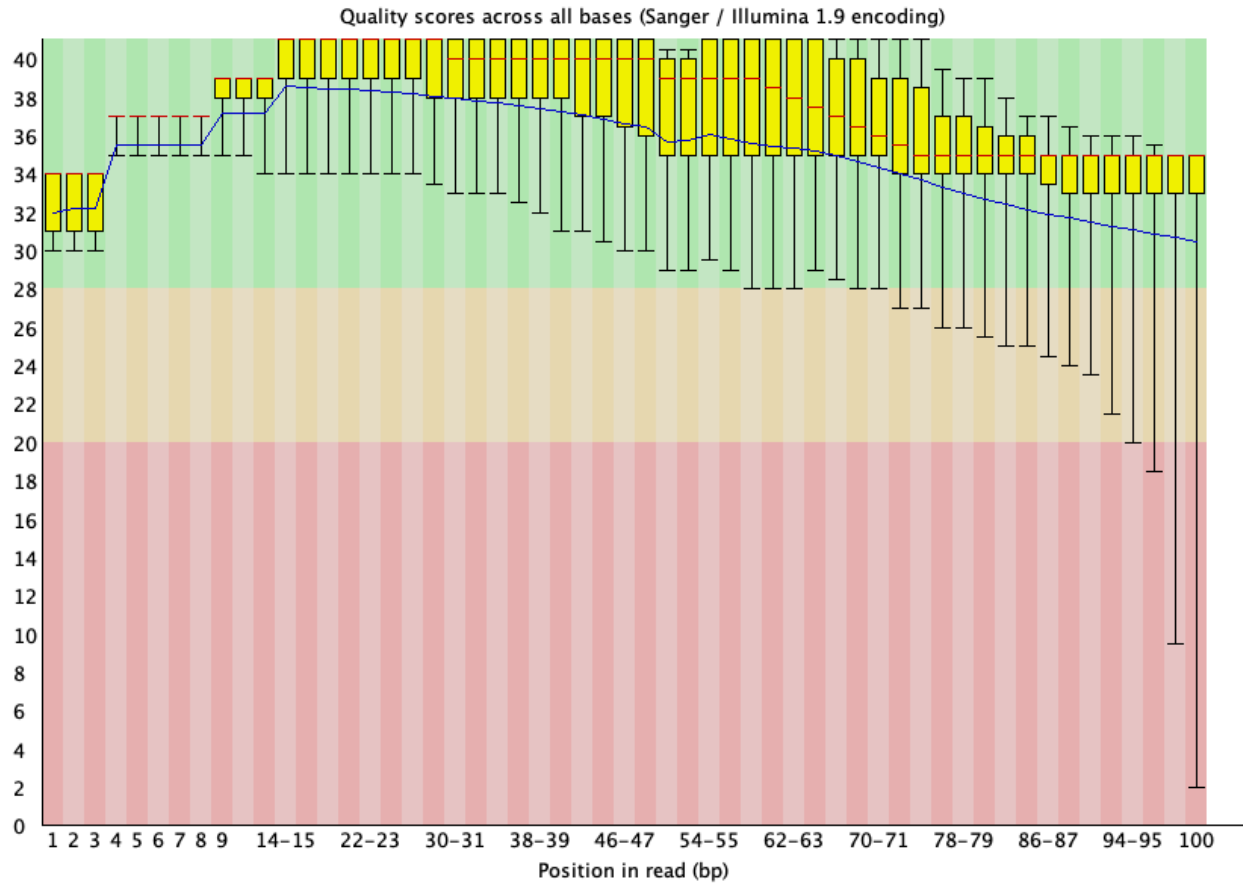
Thu 30 Jan 2025
ENCFF002DKE.fastq

Basic Statistics

Measure	Value
Filename	ENCFF002DKE.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	74096905
Total Bases	7.4 Gbp
Sequences flagged as poor quality	0
Sequence length	100
%GC	47

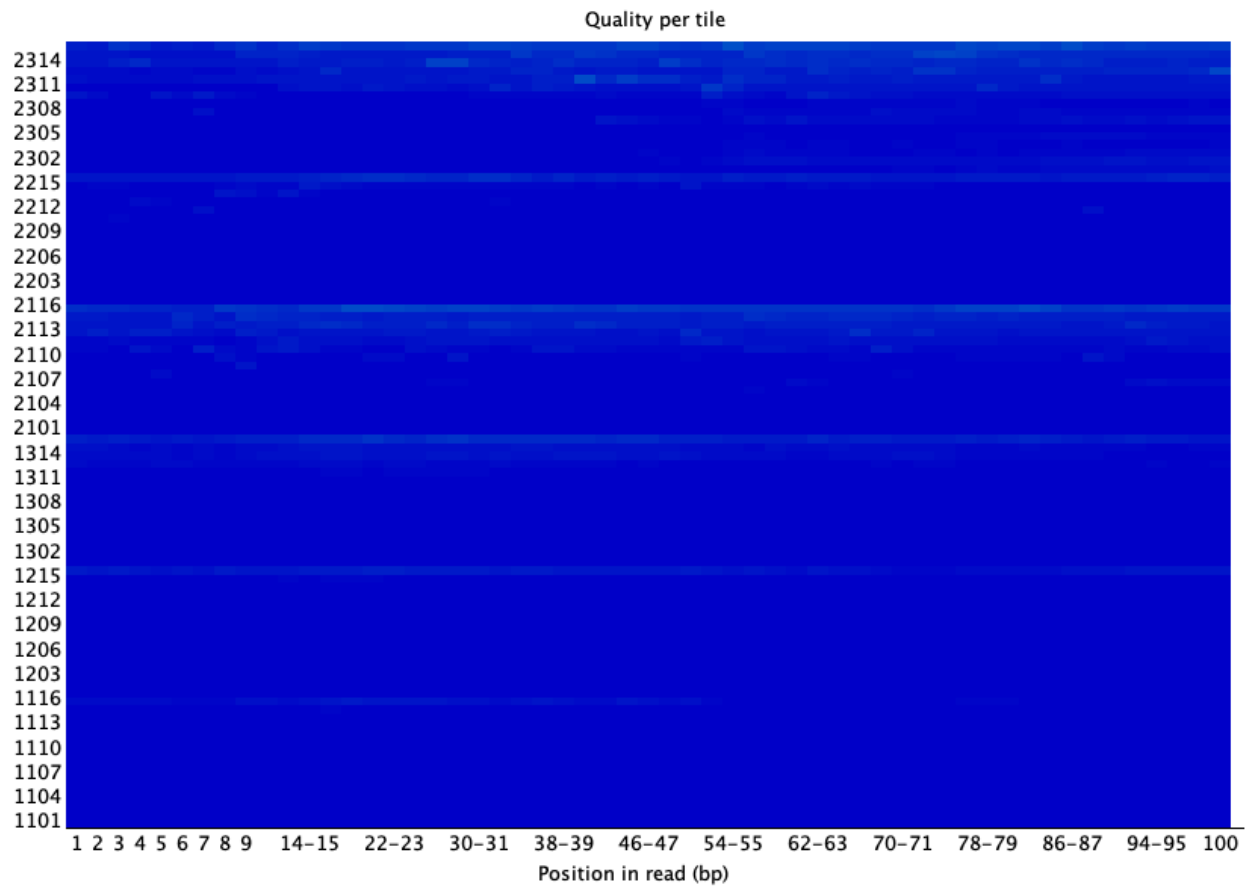
- **Description:** Provides an overview of the sequence file, including total sequences, sequence length distribution, GC content, and encoding type.
- **Graph:** A summary table showing key statistics.
- **Interpretation:** No major issues detected.

2. Per Base Sequence Quality



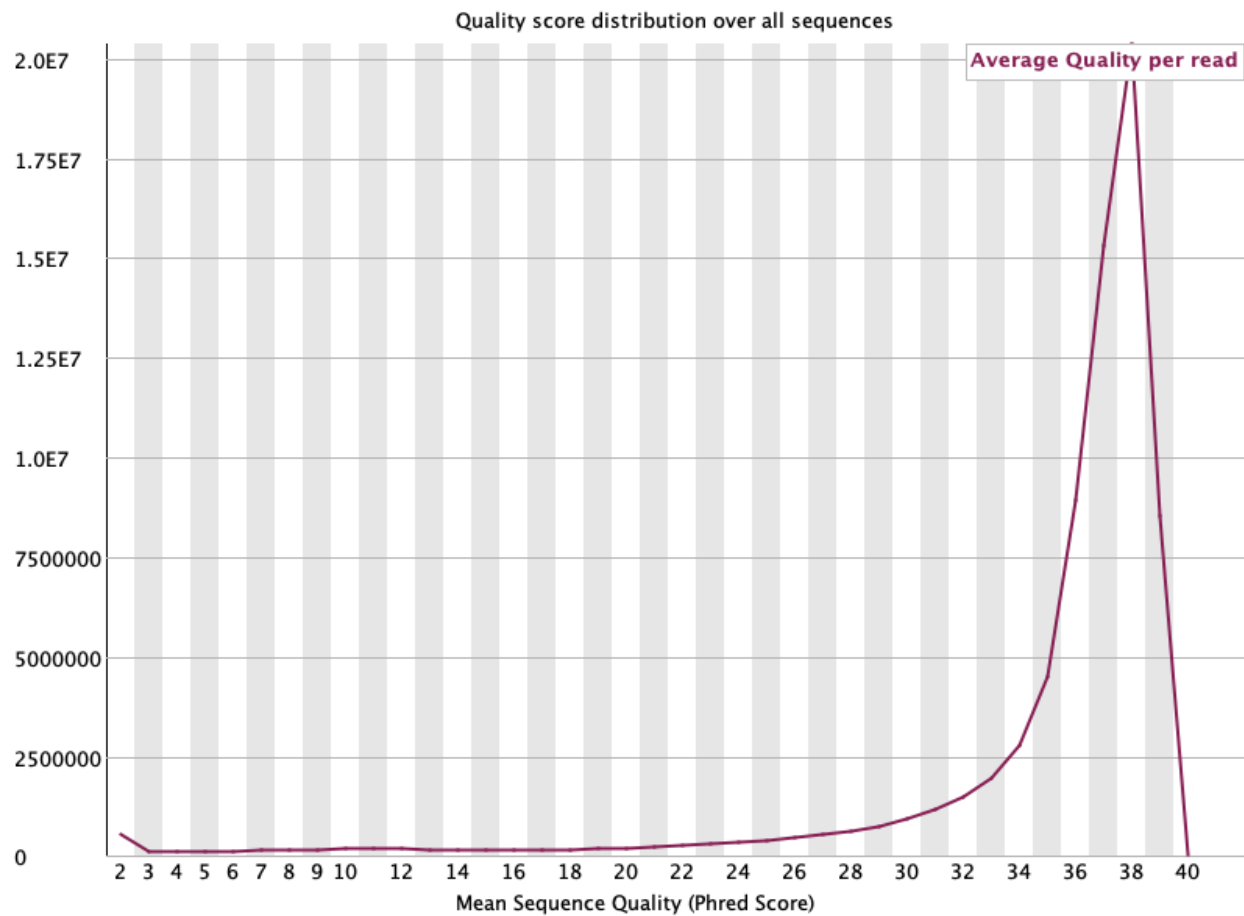
- **Description:** A box plot showing quality score distribution at each nucleotide position.
- **Graph:** Box plot with median (red line), interquartile range (yellow), and whiskers for quality variation.
- **Interpretation:** Quality slightly decreases towards the end of reads, which is expected in high-throughput sequencing.

3. Per Tile Sequence Quality



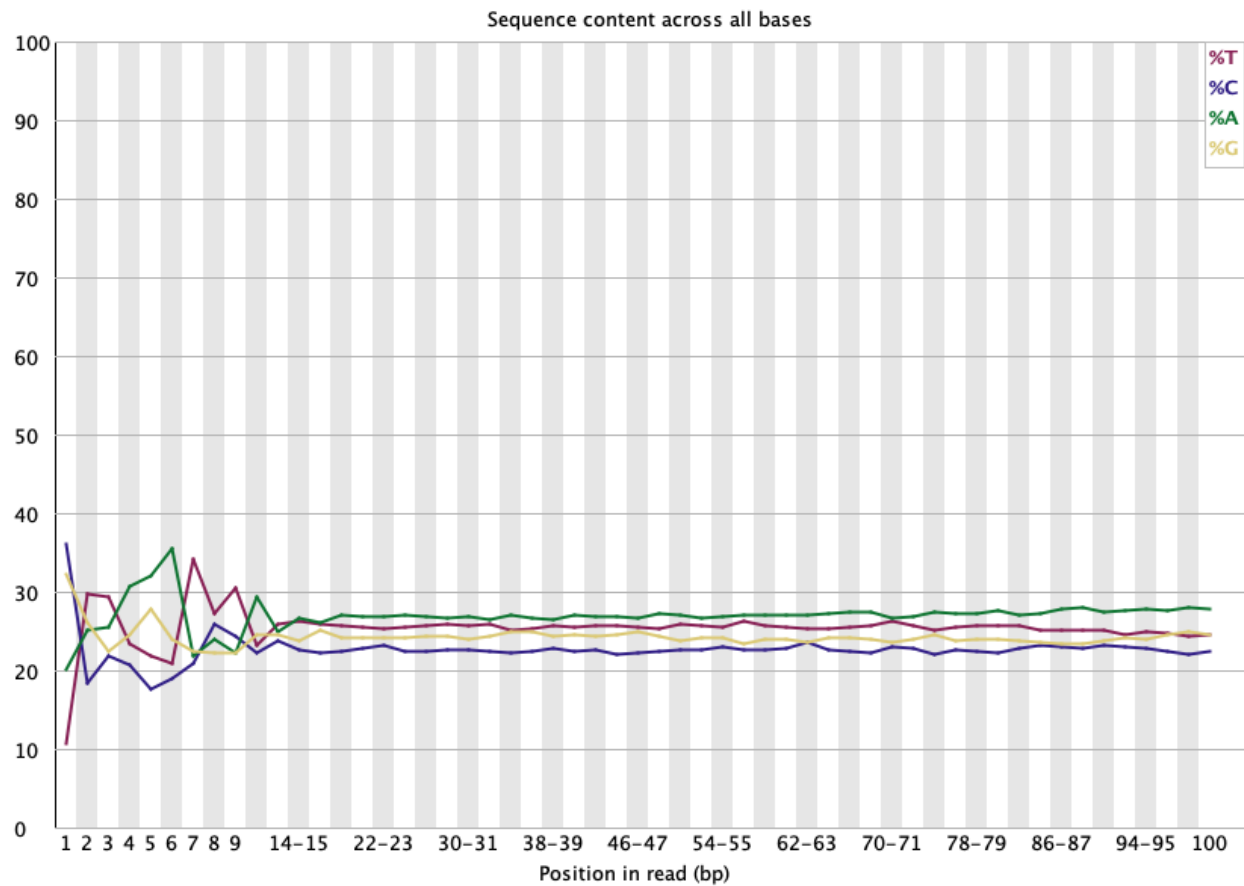
- **Description:** Identifies systematic quality loss in specific tiles of the sequencing flowcell.
- **Graph:** Heatmap with blue indicating high quality and red/yellow showing lower quality.
- **Interpretation:** No significant quality loss detected in specific tiles.

4. Per Sequence Quality Scores



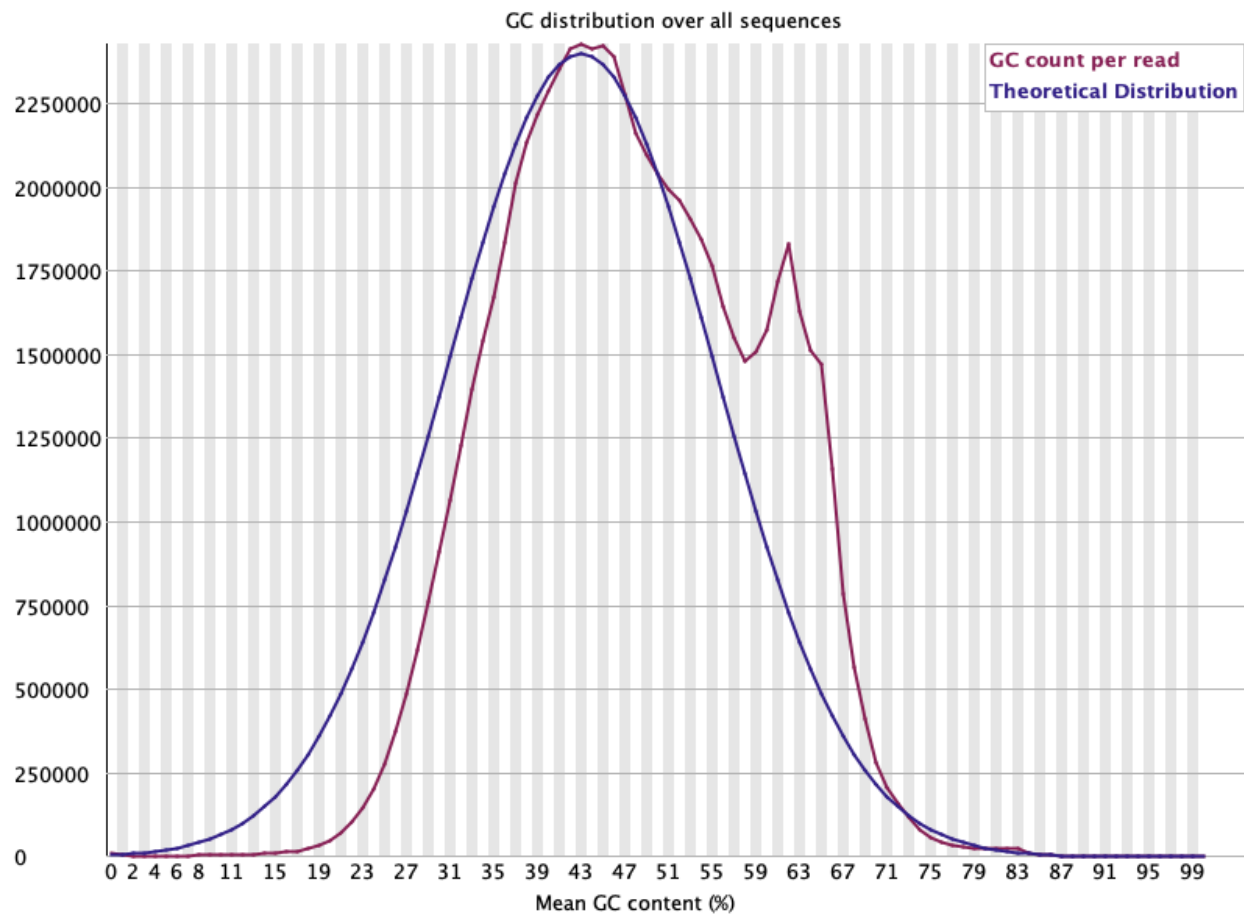
- **Description:** Histogram showing the distribution of average quality scores per read.
- **Graph:** Quality score histogram.
- **Interpretation:** Majority of sequences have high-quality scores.

5. Per Base Sequence Content



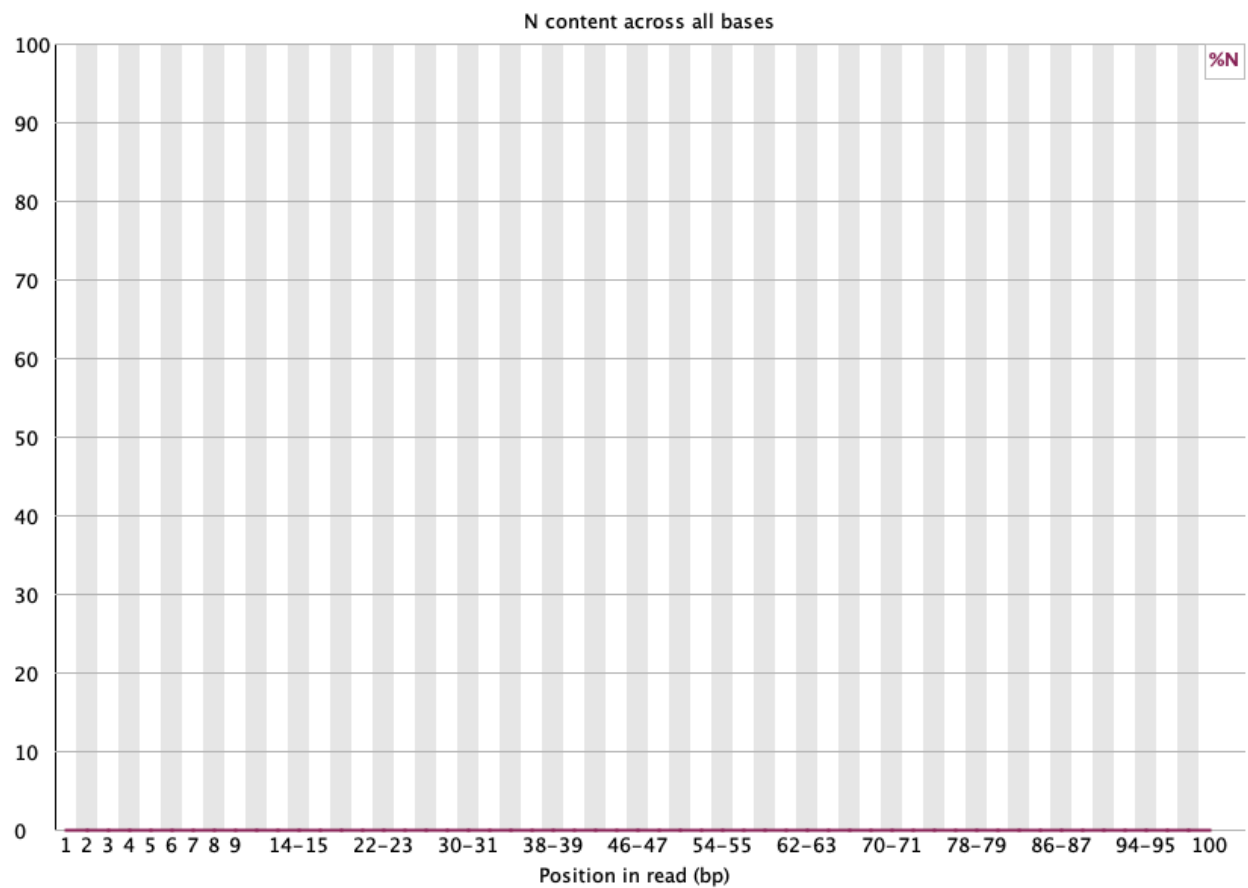
- **Description:** Tracks the proportion of A, T, G, and C bases at each position.
- **Graph:** Line plot showing percentage of each nucleotide per base position.
- **Interpretation:** Slight nucleotide bias detected, which could be due to adapter contamination.

6. Per Sequence GC Content



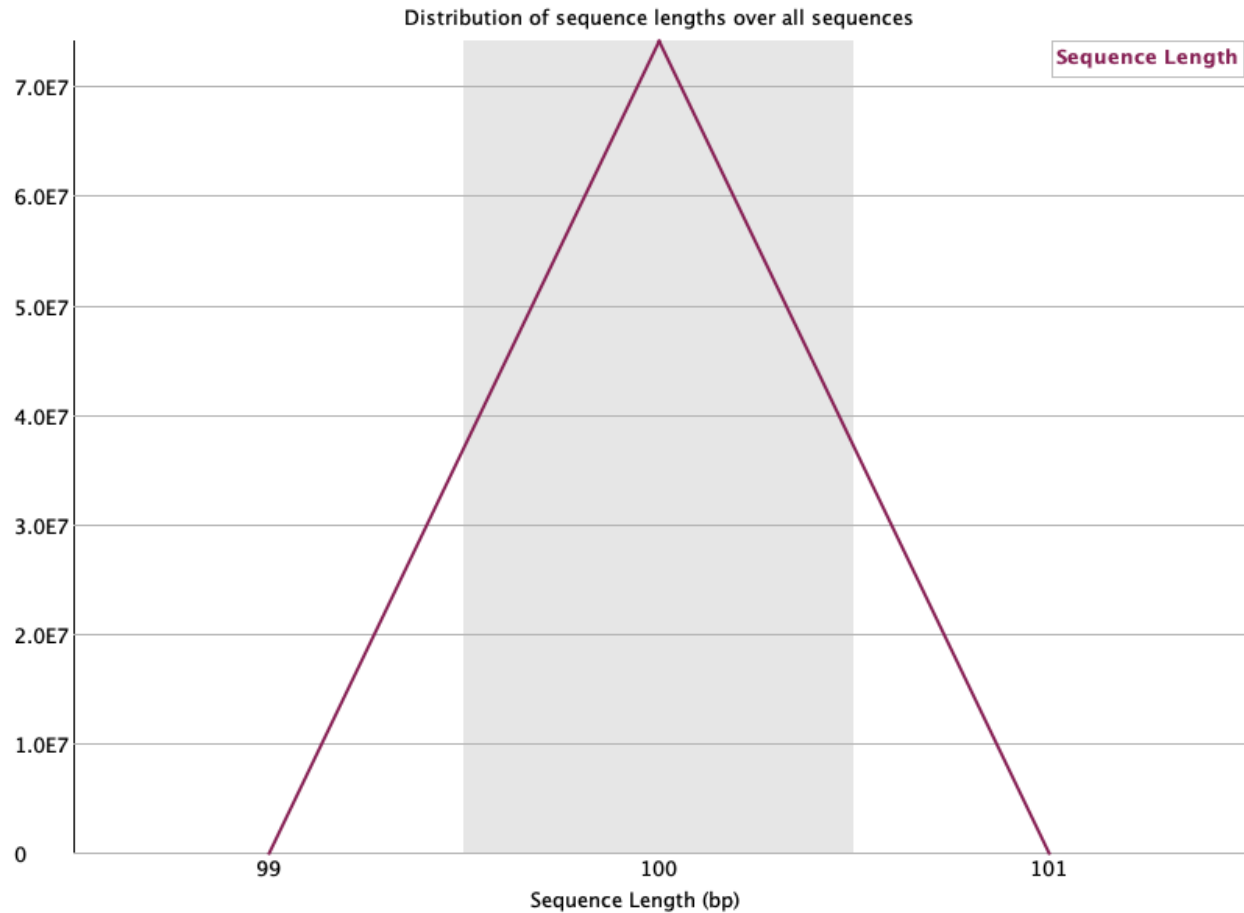
- **Description:** Compares the observed GC content distribution to an expected normal distribution.
- **Graph:** Histogram comparing expected vs. actual GC content.
- **Interpretation:** Slight deviation suggesting possible contamination or sequencing bias.

7. Per Base N Content



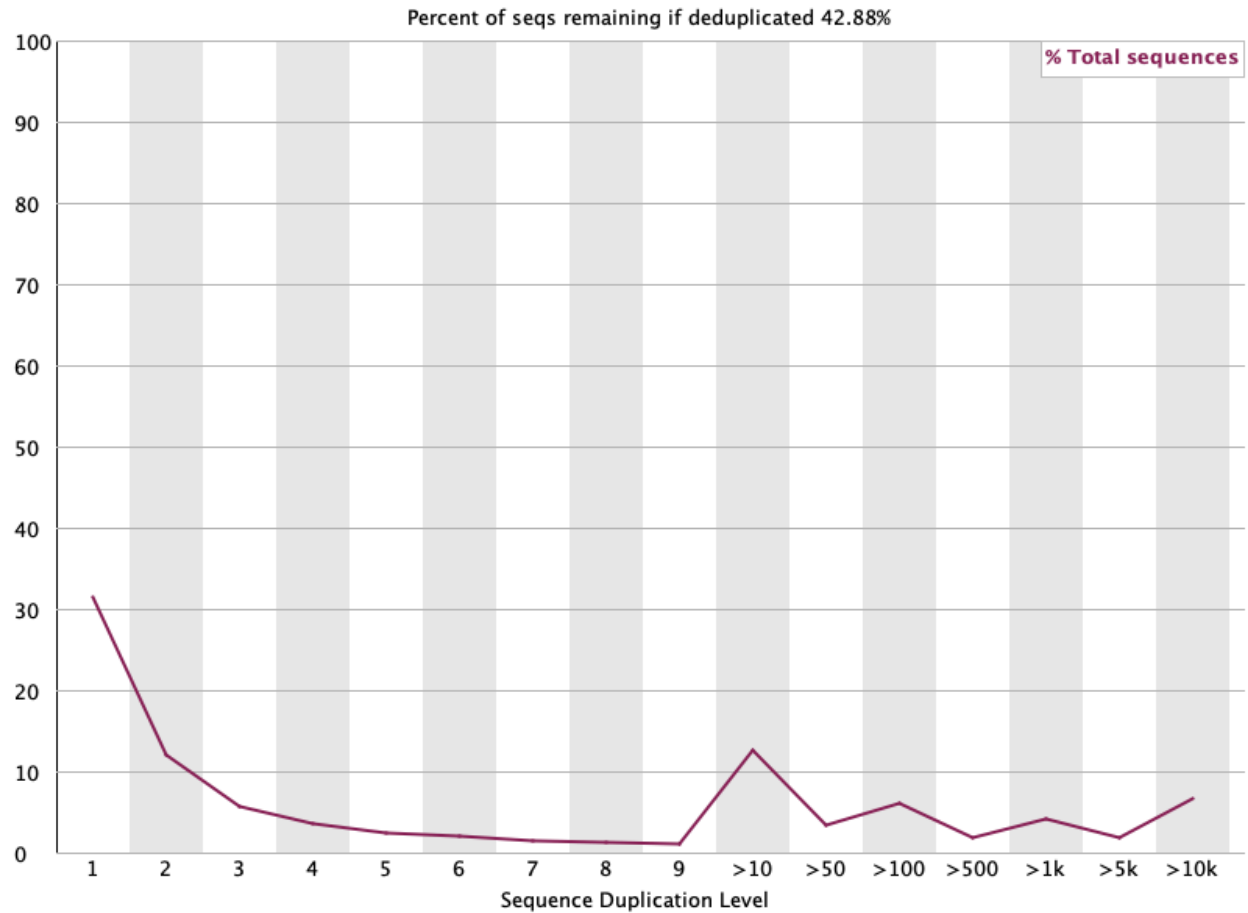
- **Description:** Measures the presence of undetermined bases (N) across positions.
- **Graph:** Line graph showing the percentage of N content at each base position.
- **Interpretation:** Minimal N content detected, indicating reliable base calling.

8. Sequence Length Distribution



- **Description:** Displays the length distribution of reads.
- **Graph:** Histogram showing the distribution of read lengths.
- **Interpretation:** Consistent read length across sequences, suggesting uniform sequencing output.

9. Sequence Duplication Levels



- **Description:** Analyzes the presence of duplicate reads.
- **Graph:** Line plot showing duplication percentage.
- **Interpretation:** Higher duplication levels detected, indicating potential PCR amplification bias.

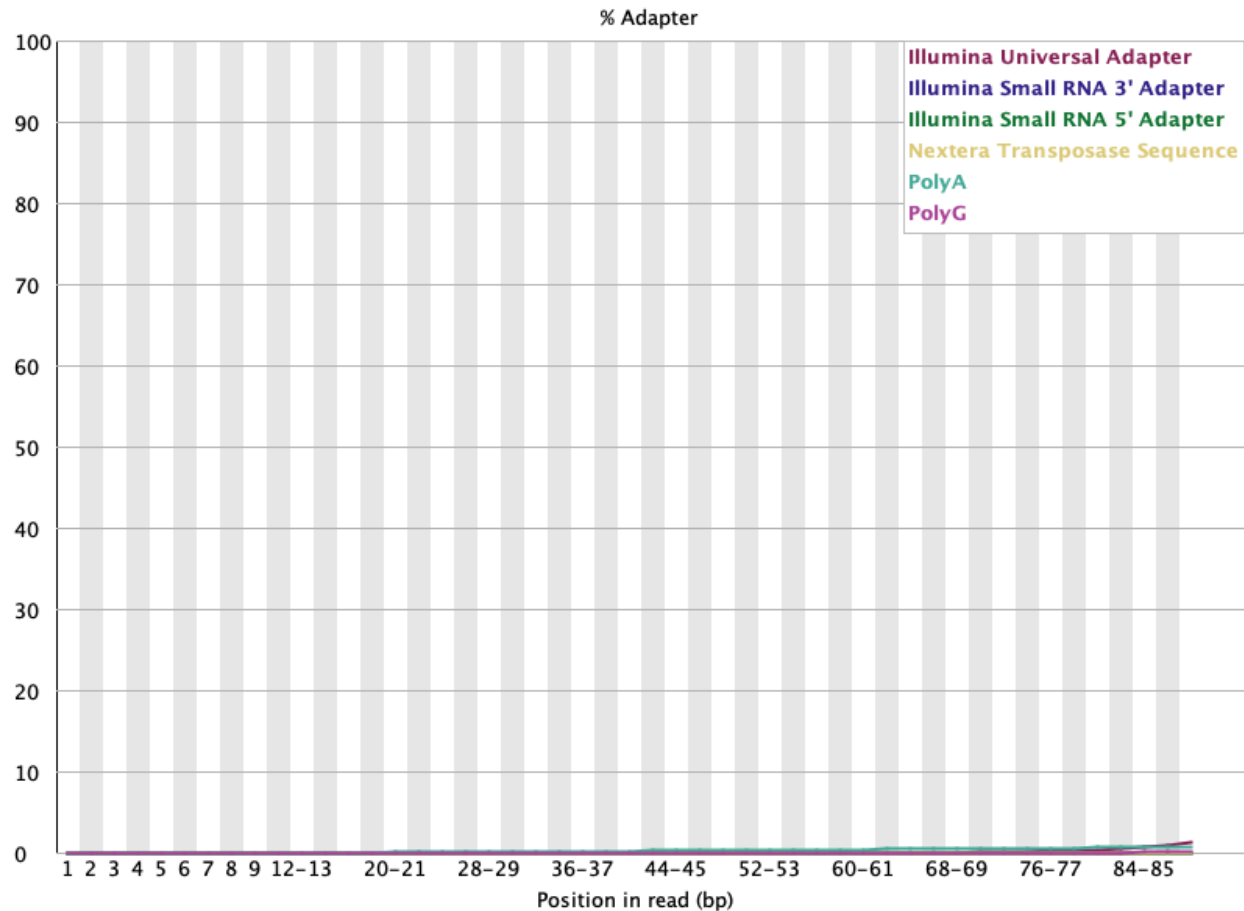
10. Overrepresented Sequences

Overrepresented sequences

Sequence	Count	Percentage	Possible Source
GGGCGATCTGGCTGCGACATCTGTCACCCCATTTGATCGCCAGGGTTGATT	206798	0.27909127918365817	No Hit
CGGTGGCGCGTGCCTGTAGTCCAGCTACTCGGGAGGCTGAGGTGGGAGG	190265	0.2567786063399004	No Hit
CGGTGGCGCGTGCCTGTAGTCCAGCTACTCGGGAGGCTGAGGTGGGAGG	187918	0.25361113261073454	No Hit
GCGGTGGCGCGTGCCTGTAGTCCAGCTACTCGGGAGGCTGAGGTGGGAG	143473	0.19362887019370106	No Hit
GCGGTGGCGCGTGCCTGTAGTCCAGCTACTCGGGAGGCTGAGGTGGGAG	140644	0.18981089695986628	No Hit
CATCTGTCACCCCATTTGATCGCCAGGGTTGATTGCGCTGATCTGGCTGGC	115438	0.15579328178417168	No Hit
CCCAGCTACTCGGGAGGCTGAGGTGGGAGGATCGCTTGAGCCAGGAGTT	114596	0.15465693202705294	No Hit
CCCAGCTACTCGGGAGGCTGAGGTGGGAGGATCGCTTGAGTCCAGGAGTT	101301	0.13671421228727434	No Hit
GGTGGCGCGTGCCTGTAGTCCAGCTACTCGGGAGGCTGAGGTGGGAGGA	97219	0.13120521025810727	No Hit
GGTGGCGCGTGCCTGTAGTCCAGCTACTCGGGAGGCTGAGGTGGGAGGA	96924	0.13080708296790536	No Hit
GTTCTGGGCTGTAGTGCCTATGCCGATCGGGTGTCGCGACTAAGTTCGG	79538	0.10734321494264842	No Hit

- **Description:** Identifies sequences appearing more frequently than expected.
- **Graph:** Table listing overrepresented sequences and their possible sources.
- **Interpretation:** Presence of overrepresented sequences, suggesting possible adapter contamination.

11. Adapter Content



- **Description:** Checks for contamination from sequencing adapters.
- **Graph:** Line graph showing the presence of adapter sequences.
- **Interpretation:** No significant adapter contamination detected.

Overall Summary:

- **Major Issues:** Nucleotide bias, high sequence duplication levels, and overrepresented sequences.
- **Minor Warnings:** Slight deviations in GC content.
- **Recommendations:** Perform adapter trimming, investigate sequence duplication sources, and assess potential contamination sources.

This report provides a comprehensive visual analysis of sequencing quality, aiding in data preprocessing for downstream bioinformatics analysis.